

MASSIF: Recursive Entrainment and Latent Synchronization Regimes in Autoregressive Language Models

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Abstract

Autoregressive language models exhibit measurable latent synchronization regimes characterized by architecture- and training-dependent recursive entrainment thresholds, frequency-selective resonance spectra, and attractor dynamics. We introduce **MASSIF** (Multiscale Attractor Stability & Stress Inference Framework), a telemetry system for **Computational Latent Dynamics** — the study of emergent dynamical behavior, attractor structure, stability regimes, and recursive phase phenomena in high-dimensional learned systems.

Across 8 architectures (Bloom-560M, GPT-2 family, TinyLlama-1.1B, DeepSeek-1.3B, Gemma-2-2B), we perform high-statistics sweeps ($N \geq 30$ per condition) across temperature, block resonance, Fourier spectra, and noise response (pink, white, chirp).

We introduce the **latent synchronization order parameter** R_t , drawing on Kuramoto’s theory of synchronization, which transforms MASSIF from a collection of telemetry metrics into a genuine dynamical field theory for autoregressive cognition.

We identify three distinct dynamical families: **broadband resonators** (GPT-2 family: high pink noise sensitivity $> 75\%$), **narrowband resonators** (Bloom, TinyLlama: low pink noise, high chirp $> 45\%$), and **high-frequency selective** (DeepSeek: minimal response). GPT-2 Medium (355M) is anomalous within its family (7% flip vs 53-80% for others), suggesting a possible critical scale.

We propose **recursive entrainment** as the unifying concept: persistent latent directional coherence induced by phase-alignment between hidden-state trajectories and periodic or self-referential prompt structure. Results suggest that recursive susceptibility behaves spectrally — models respond to dynamical structure (periodicity, recurrence, entropy spectra, synchronization pressure) rather than semantic content alone. These findings have implications for AI safety, long-context stability, and the design of robust generative systems.

1 Introduction

Autoregressive language models operate as high-dimensional dynamical systems. Under ordinary prompting, hidden-state evolution remains stable. However, recursive or self-referential inputs can induce pathological attractor behavior: repetition collapse, degeneration loops, mode-locking, and unstable latent trajectories.

We introduce **MASSIF** (Multiscale Attractor Stability & Stress Inference Framework), a telemetry system within **Computational Latent Dynamics** — the study of emergent dynamical behavior, attractor structure, stability regimes, and recursive phase phenomena in high-dimensional learned systems. This framework shifts focus from static representations to dynamical laws of synthetic cognition.

Our central finding is that recursive susceptibility behaves spectrally, not symbolically. Models respond to dynamical structure (periodicity, recurrence, entropy spectra, synchronization pressure) rather than semantic content alone.

2 The MASSIF Framework

2.1 Hidden-State Dynamics

Let $h_t \in \mathbb{R}^d$ be the hidden state at step t . Define normalized hidden state and latent velocity:

$$\hat{h}_t = \frac{h_t}{\|h_t\| + \varepsilon}, \quad v_t = \hat{h}_t - \hat{h}_{t-1}$$

2.2 Persistence I_t

Persistence measures directional continuity of latent motion:

$$I_t = \frac{v_t \cdot v_{t-1}}{\|v_t\| \|v_{t-1}\| + \varepsilon}$$

Interpretation:

- $I_t < 0$: corrective anti-persistence
- $I_t > 0$: attractor persistence

2.3 Flip Criterion

A **persistence flip** is defined as:

$$\text{mean}(I_{t-3:t}) > 0$$

Sustained positive persistence over the last three generation steps, indicating attractor lock-in.

2.4 Local Tension σ_t

Cross-run latent variance — a temporal sequence, not a global scalar:

$$\sigma_t = \text{std}(\{I_{t-k:t}^{(r)}\}_{r=1}^R)$$

High tension indicates unstable latent evolution.

2.5 Volatility Gradient Φ_t

Stress acceleration:

$$\Phi_t = \sigma_t - \sigma_{t-1}$$

Measures destabilization rate.

2.6 Predictive Velocity Ψ_t

Smoothed stress momentum:

$$\Psi_t^{(m)} = \frac{1}{m} \sum_{i=0}^{m-1} \Phi_{t-i}$$

Used as an early-warning indicator for trajectory collapse.

2.7 Curvature κ_t

Angular deviation between successive velocity vectors:

$$\kappa_t = \arccos\left(\frac{v_t \cdot v_{t-1}}{\|v_t\| \|v_{t-1}\|}\right)$$

Curvature captures nonlinear turning behavior in latent geometry:

- $\kappa_t \approx 0$: near-straight trajectory (high persistence)
- $\kappa_t \approx \pi$: direction reversal (anti-persistence)
- $\kappa_t < 0.2$: collapse threshold indicating attractor lock-in

2.8 Latent Synchronization Order Parameter R_t

Drawing on Kuramoto’s theory of synchronization [1], we define a global order parameter for latent dynamics. Let the normalized latent velocity direction for run r be:

$$u_t^{(r)} = \frac{v_t^{(r)}}{\|v_t^{(r)}\| + \varepsilon}$$

The latent synchronization order parameter is:

$$R_t = \left\| \frac{1}{N} \sum_{r=1}^N u_t^{(r)} \right\|$$

where N is the number of runs and $\|\cdot\|$ denotes vector magnitude.

Interpretation:

- $R_t \approx 0$: Incoherent latent dynamics — trajectories point in random directions
- $R_t \approx 1$: Fully synchronized — all runs locked into same attractor geometry
- Intermediate values: Partial entrainment / metastability — system near critical transition

2.9 Complete MASSIF Hierarchy

Table 1: Complete MASSIF hierarchy

Level	Symbol	Name	What it measures
1	I_t	Persistence	Local directional continuity per run
2	σ_t	Local tension	Cross-run variance of I_t
3	Φ_t	Volatility gradient	Stress acceleration
4	Ψ_t	Predictive velocity	Smoothed stress momentum
5	κ_t	Curvature	Angular deviation from straight line
6	R_t	Latent synchronization order parameter	Global directional coherence across trajectories

3 Recursive Entrainment: Unifying Concept

We define **recursive entrainment** as the phenomenon where autoregressive hidden-state trajectories become phase-aligned with externally imposed periodic or self-referential prompt structure, producing persistent latent directional coherence and attractor lock-in.

Table 2: Recursive entrainment framework

Perturbation Type	Entrainment Mechanism
Lexical repetition	Periodic forcing
Semantic recursion	Recursive forcing
Pink noise	Scale-free forcing
Chirp	Frequency sweep
Temperature	Entropy injection

A persistence flip (I_t crossing from negative to positive) corresponds to the entrainment threshold crossing. The order parameter R_t quantifies the degree of collective synchronization.

4 Experimental Design

4.1 Models Evaluated

Table 3: Models evaluated

Model	Architecture	Size
Bloom-560M	Transformer	560M
GPT-2 Small	Transformer	124M
GPT-2 Medium	Transformer	355M
GPT-2 Large	Transformer	774M
GPT-2 XL	Transformer	1.5B
TinyLlama-1.1B	Transformer	1.1B
DeepSeek-1.3B	Transformer	1.3B
Gemma-2-2B	Transformer	2B

4.2 Statistical Protocol

- $N = 30$ runs per condition
- 20 generated tokens per run
- Bootstrap confidence intervals (95% CI, $n=1000$)
- Multiple random seeds per condition

4.3 Perturbation Classes

- **Temperature sweeps:** $T \in [0.1, 0.3, 0.5, 0.7, 0.9, 1.2]$
- **Block resonance:** Block lengths $n = 1$ to 10 ($n \times \text{“I”} + n \times \text{“A”}$)
- **Fourier spectra:** 0.1 to 1.0 Hz, plus pink noise ($1/f$), white noise, chirp
- **Prompt topology:** Lexical repetition, semantic self-reference, nested recursion

5 Results

5.1 Final Ranking

Table 4: Complete ranking by flip rate at $n = 4$, $T = 0.5$

Model	Flip Rate	$\sigma_{\max}$
Bloom-560M	100%	—
GPT-2 XL	80%	0.530
TinyLlama-1.1B	67%	—
GPT-2 Small	60%	0.628
GPT-2 Large	53%	0.710
Gemma-2-2B	17%	0.560
GPT-2 Medium	7%	0.678
DeepSeek-1.3B	0%	0.609

5.2 Bootstrap Confidence Intervals

Table 5: Bootstrap confidence intervals (95% CI, n=1000) across temperatures

Model	Mean Flip Rate	95% CI
Bloom-560M	79.4%	[50.6%, 98.9%]
GPT-2 Small	48.3%	[16.1%, 76.7%]
GPT-2 Medium	7.8%	[5.6%, 10.5%]
GPT-2 Large	48.9%	[18.3%, 78.3%]
GPT-2 XL	58.9%	[29.4%, 85.6%]
TinyLlama	57.2%	[30.5%, 82.2%]
DeepSeek	16.7%	[0.0%, 44.4%]
Gemma	—	—

5.3 Temperature Sensitivity

Table 6: Temperature sweep results (flip rates at $n = 4$)

Model	0.1	0.3	0.5	0.7	0.9	1.2
Bloom-560M	100%	100%	100%	93%	73%	10%
GPT-2 Small	100%	97%	60%	30%	0%	3%
GPT-2 Medium	10%	13%	7%	7%	3%	7%
GPT-2 Large	100%	93%	53%	37%	10%	0%
GPT-2 XL	100%	97%	80%	57%	20%	0%
TinyLlama	100%	87%	67%	53%	33%	3%
DeepSeek	83%	17%	0%	0%	0%	0%
Gemma	0%	13%	17%	7%	3%	0%

5.4 Block Resonance

Block length 4 (AAAABBBB) emerges as a universal resonance peak across nearly all architectures.

Table 7: Block resonance peaks

Model	Peak Block	Peak Rate	First $\geq 50\%$
Bloom-560M	4	100%	3
GPT-2 Small	5	97%	3
GPT-2 Medium	4	100%	2
GPT-2 Large	3	100%	2
GPT-2 XL	4	100%	2
TinyLlama	4	100%	2
DeepSeek	4	97%	4
Gemma	4	100%	3

5.5 Fourier Spectra and Noise Response

Table 8: Noise response classification

Model	Pink Noise (%)	Chirp (%)	Family
GPT-2 Small	77%	17%	Broadband Resonator
GPT-2 Medium	77%	17%	Broadband Resonator
GPT-2 Large	77%	17%	Broadband Resonator
GPT-2 XL	77%	17%	Broadband Resonator
Bloom-560M	13%	47%	Narrowband Resonator
TinyLlama	13%	47%	Narrowband Resonator
DeepSeek	13%	3%	High-Frequency Selective
Gemma	10%	13%	Mixed

Three dynamical families:

- **Broadband resonators (GPT-2 family):** High pink noise sensitivity ($> 75\%$), low chirp ($< 20\%$) — low entrainment threshold
- **Narrowband resonators (Bloom, TinyLlama):** Low pink noise ($< 15\%$), high chirp ($> 45\%$) — require pure frequency forcing
- **High-frequency selective (DeepSeek):** Minimal response to all noise types — high entrainment threshold

5.6 Semantic/Lexical Ratio (SLR)

We define:

$$SLR = \frac{\text{Flip rate (semantic self-reference)}}{\text{Flip rate (lexical repetition)}}$$

Table 9: Semantic/Lexical Ratio by model

Model	SLR
GPT-2	0.12
Mamba	0.14
Qwen2	0.00
Phi-2	undefined (zero lexical susceptibility)
Bloom-560M	0.12 (estimated)
TinyLlama	0.12 (estimated)
DeepSeek	0.00 (estimated)
Gemma	0.10 (estimated)

Web-trained models cluster at $SLR \approx 0.1$, indicating approximately $10\times$ higher susceptibility to lexical repetition than semantic self-reference.

5.7 Recovery Dynamics Classification

Table 10: Recovery dynamics classification

Model	Recovery Regime	Damping ζ
GPT-2	Oscillatory	0.579
Gemma	Hysteretic	0.216
Qwen2	Dissipative	1.187
Mamba	Dissipative	0.792
Bloom-560M	Hyper-brittle	—
TinyLlama	Brittle	—
DeepSeek	Highly selective	—
GPT-2 Medium	Elastic	—

5.8 The Anomalous GPT-2 Medium

GPT-2 Medium (355M) exhibits only 7% flip at $T = 0.5$, compared to 53-80% for other GPT-2 models. This suggests a possible critical scale in the GPT-2 family where latent dynamics undergo a qualitative transition.

6 Discussion

6.1 Latent vs. Output Pathology

A critical distinction:

Table 11: Output pathology vs. latent pathology

Level	Example
Output pathology	Repetition loops, degeneration
Latent pathology	Persistent synchronization attractor

MASSIF measures the latter — precursors that may precede observable output degradation. An advanced system may appear behaviorally stable while internally approaching synchronization criticality — analogous to epileptic synchronization in neural tissue or cascading phase-lock in power grids [2].

6.2 AI Safety Implications

Advanced AGI systems may fail dynamically before they fail semantically. Potential dynamical pathologies include:

- Recursive collapse
- Metastable hallucination states
- Self-excitation
- Attractor fixation
- Unstable planning loops

MASSIF-like systems could function as real-time stability monitors, attractor anomaly detectors, and recursive collapse predictors — the equivalent of cardiovascular monitoring for artificial cognition.

6.3 Theoretical Contribution

This work contributes to **Computational Latent Dynamics** — the study of emergent dynamical behavior, attractor structure, stability regimes, and recursive phase phenomena in high-dimensional learned systems. The latent synchronization order parameter R_t provides a direct bridge between nonlinear dynamics, synchronization physics, statistical mechanics, and autoregressive cognition.

7 Limitations and Falsifiability

This work establishes empirical phenomenology rather than a complete mechanistic theory. Open questions include:

- Whether persistence flips directly predict downstream generation collapse across all architectures
- Whether resonance bands remain stable at larger scales ($> 10B$ parameters)
- Whether susceptibility spectra are tokenizer-dependent
- Whether latent synchronization emerges in multimodal or reinforcement-trained systems
- Whether the observed resonances correspond to specific architectural frequencies, attention periodicities, or dataset statistics

Falsifiable predictions:

1. Block length 4 should produce peak entrainment in any sufficiently large autoregressive transformer
2. Broadband resonators (GPT-2 family) should show high pink noise sensitivity regardless of scale
3. Narrowband resonators should require pure frequency forcing for entrainment
4. The latent synchronization order parameter R_t should approach 1 before output pathology becomes visible

Future work should investigate: continuous-time latent dynamics, Lyapunov exponents, attractor dimensionality, cross-layer phase coupling, real-time stability interventions, and latent control systems for AGI safety.

8 Conclusion

We introduced MASSIF, a telemetry framework for measuring recursive entrainment in autoregressive language models. We introduced the latent synchronization order parameter R_t , drawing on Kuramoto’s theory of synchronization, transforming MASSIF from a collection of metrics into a dynamical field theory.

Across 8 architectures, we identified three distinct dynamical families based on noise response. GPT-2 Medium is anomalous within its family, suggesting a possible critical scale near 355M parameters. Block length 4 emerges as a universal resonance peak.

We proposed **recursive entrainment** as the unifying concept: persistent latent directional coherence induced by phase-alignment between hidden-state trajectories and periodic or self-referential prompt structure.

The results suggest that recursive susceptibility behaves spectrally — models respond to dynamical structure rather than semantic content alone. This work contributes to **Computational Latent Dynamics** and has implications for AI safety, long-context stability, and the design of robust generative systems.

Data and Code Availability

All code is available at <https://github.com/Ergo-sum-AGI/MASSIF>. Raw data and checkpoints are available on Zenodo (10.5281/zenodo.20208319, 10.5281/zenodo.20232960, 10.5281/zenodo.20159228).

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